

Diabetes Prediction Based on Limited Medical Indication

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ABSTRACT

Using machine learning methods based on relevant physiological indicators for disease prediction has become a relatively mature and widely applied technique. Diabetes is a common and prevalent disease, posing a significant global public health problem that seriously threatens human health. Currently, various machine learning and deep learning techniques, such as neural networks, have been employed for diabetes prediction. Most methods have achieved significant results by utilizing complex models on large datasets. However, many discussions on these methods lack classification analysis of experimental datasets or unilaterally pursue high precision. In the case of potential patients, a model with high recall rate becomes more important, as it reduces the probability of potential patients being misclassified as normal individuals. In this study, detailed analysis of various attributes of the dataset was conducted through a hybrid approach based on statistics and mathematics. Based on the conclusions drawn from this stage and considering factors of diabetes in society, the composition of potential cases in a small dataset was analysed. The performance of common machine learning models was tested in experiments. Ultimately, a linear model was selected and optimized, and model performance was evaluated using the confusion matrix and ROC curve, demonstrating balanced and satisfactory precision and recall scores.

CCS CONCEPTS

• **Information systems** → Information systems applications; Data mining; • **Computing methodologies** → Machine learning; Learning paradigms; Supervised learning.

KEYWORDS

Diabetes prediction, mathematical and statistical methods, machine learning

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1 INTRODUCTION

Diabetes is a metabolic disorder characterized by chronic hyperglycemia, which is caused by a combination of genetic and environmental factors. With the progression of the disease, multiple system damages can occur, including the eyes, kidneys, nerves, heart, and blood vessels. In severe cases or during periods of stress, acute metabolic disorders such as ketoacidosis and hyperosmolar hyperglycemic syndrome may also occur.

This study aims to develop a machine learning model using pathological features extracted from diabetic patients to predict the presence of diabetes based on corresponding physiological characteristics. The dataset was initially partitioned based on priori knowledge. Subsequently, we conducted distribution analysis on the continuous attributes and applied appropriate preprocessing techniques. Various models, including Support Vector Machines, Decision Trees, K-Nearest Neighbors, Linear Regression, and ensemble methods like Random Forests, were evaluated using k-fold cross-validation on the training set. Ultimately, a linear regression model was selected for prediction on the dataset. Hyperparameter optimization was performed to enhance the performance of the model.

2 RELATED WORKS

In recent years, there have been various studies utilizing machine learning or deep learning methods to analyze and predict diabetes based on Pima Indian Dataset same as our experiment.

One relevant work by Yang Guo et al. [1] focused on using a Bayes Network to predict the development of Type-2 diabetes. They constructed a naïve Bayes model and achieved improved accuracy, reaching up to 72.3%.

Following that, M. NirmalaDevi et al. [2] conducted diabetes prediction work utilize the K-Nearest Neighbors (KNN) algorithm. They employed amalgam KNN and achieved an impressive precision of 97.4% on the dataset through 10-fold cross-validation when k was set to 5. However, it should be noted that KNN-based approaches tend to incur higher computational complexity when dealing with larger datasets. Additionally, the generalizability of KNN-based methods can be relatively weaker in the presence of imbalanced data or when dealing with cases that are not well-represented in the dataset.

N. Nnamoko et al. [3] employed ensemble learning methods, combining five base classifiers: Sequential Minimal Optimization (SMO), Radial Basis Function (RBF), C4.5 decision tree, Naïve Bayes (NB), and Repeated Incremental Pruning to Produce Error Reduction (RIPPER). The experiment ultimately achieved a high accuracy of 83%. Their results demonstrated that improvements were achieved through feature selection. Similarly, G. Verma et al. [4] utilized the similar base classifiers and obtained similar accuracy through a voting mechanism on the original dataset.